

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA0706

COURSE OUTCOME COVERED

CO2: *Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)*

Assessment Tool Weightage: 30%

Annexure A

Scenario-Based

Code of Conduct:

I S. Karthikeyan / 192419048 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

CSA0706—COMPUTER NETWORKS

Assessment Tool 2—Scenario-Based Questions: Answers

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques for efficient network design

Q1. Smart Campus—Variable-Length Subnetting on 192.168.100.0/24

Allocated block: **192.168.100.0/24** (256 addresses). Assigned masks: School of Computing **/25**, School of Electronics **/26**, School of Mechanical Engineering **/27**. Since the departments need different-sized subnets from one block, this is solved with **Variable Length Subnet Masking (VLSM)**—allocating the largest requirement first so each subnet's block stays properly aligned.

VLSM Allocation

Remaining Unused Address Range

Department	Subnet	Network Address	Broadcast Address	Valid Host Range	Usable Hosts
School of Computing	/25	192.168.100.0	192.168.100.127	192.168.100.1–.126	126
School of Electronics	/26	192.168.100.128	192.168.100.191	192.168.100.129–.190	62
School of Mechanical Eng.	/27	192.168.100.192	192.168.100.223	192.168.100.193–.222	30

The three allocated subnets together consume 192.168.100.0 through 192.168.100.223 (128 + 64 + 32 = 224 addresses out of 256). The remaining, still-unused block is:

- 192.168.100.224–192.168.100.255 (a full /27 block of 32 addresses).

Q2. Multinational Software Company—VLSM on 10.10.0.0/16

Allocated block: **10.10.0.0/16** (65,536 addresses). Assigned masks: Software Development **/24**, Quality Assurance **/25**, Human Resources **/26**, Executive Management **/27**. Allocating sequentially, largest block first, keeps every subnet correctly aligned to its own size.

VLSM Allocation

Department	Subnet	Network Address	Broadcast Address	Valid Host Range	Usable Hosts
Software Development	/24	10.10.0.0	10.10.0.255	10.10.0.1–.254	254
Quality Assurance	/25	10.10.1.0	10.10.1.127	10.10.1.1–.126	126
Human Resources	/26	10.10.1.128	10.10.1.191	10.10.1.129–.190	62
Executive Management	/27	10.10.1.192	10.10.1.223	10.10.1.193–.222	30

Remaining Unused Address Range

The four subnets together use $256 + 128 + 64 + 32 = 480$ addresses out of the 65,536 available in the /16 block. The remaining unused range is:

- 10.10.1.224–10.10.255.255 (65,056 addresses remain free for future departments or growth).

Q3. Smart Hospital—IPv4-to-IPv6 Migration

Assigned IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000/64

1. Compressed Notation

2001:db8:abcd:: —leading zeros are dropped from the "0db8" group(→db8), and the run of five consecutive all-zero groups is replaced once with "::".

2. Expanded (Full Hexadecimal) Notation

2001:0db8:abcd:0000:0000:0000:0000:0000 —every group written out to 4 hex digits, with "::" restored to explicit "0000" groups.

3. Network Prefix (/64)

The first 64 bits (the first four groups) form the network portion:

2001:0db8:abcd:0000::/64 (equivalently written **2001:db8:abcd::/64**).

4. Total Host Addresses in the /64 Subnet

Remaining host bits = $128 - 64 = 64$ bits → total possible host addresses = $2^{64} =$ **18,446,744,073,709,551,616** addresses—comfortably enough to individually address every medical device and IoT sensor the hospital could ever deploy, which is precisely why IPv6's /64 subnet convention was chosen for this migration.

Q4. ISP Core Router—Routing Table Lookup

Routing table entries: 192.168.1.0/24, 192.168.2.0/24, 192.168.3.0/24. Incoming packet destination:

192.168.2.45

1. Destination Subnet

192.168.2.45 lies within the range 192.168.2.0–192.168.2.255, so it belongs to the

192.168.2.0/24 subnet.

2. Matching Routing Table Entry

The router performs a longest-prefix match against each entry; only **192.168.2.0/24** matches the destination address 192.168.2.45 (the /24 mask covers 192.168.2.0–192.168.2.255, which includes .45; the other two entries, 192.168.1.0/24 and 192.168.3.0/24, do not include this address).

3. Next-Hop Network / Interface

The packet is forwarded out of whichever router interface is configured for the 192.168.2.0/24 network in the routing table (typically shown in a real routing table as either "directly connected" for that interface, or a specific next-hop IP address on that link)—i.e., the interface serving the customer network whose routing entry matched, **192.168.2.0/24**.

4. Default Route (No Match Case)

If an incoming destination address did not match any of the three specific /24 entries, the router would forward it using its **default route**, conventionally represented as

0.0.0.0/0, which points to the ISP's upstream gateway/interface for any traffic.

Q5. Two-Department Office—Layer-3 Routed LANs

Administration LAN: **192.168.10.0/26** (gateway 192.168.10.1). Finance LAN: **192.168.10.64/26** (gateway 192.168.10.65).

1 & 2. Network/Broadcast Address and Valid Host Rang

3. Suggested IP Addresses for Three Computers per LAN

LAN	Subnet	Network Address	Broadcast Address	Valid Host Range	Usable Hosts
Administration	/26	192.168.10.0	192.168.10.63	192.168.10.1–.62	62
Finance	/26	192.168.10.64	192.168.10.127	192.168.10.65–.126	62

(Any address inside the valid host range, excluding the address already used by the gateway, is suitable.)

LAN	PC 1	PC 2	PC 3
Administration (gateway .1)	192.168.10.2	192.168.10.3	192.168.10.4
Finance (gateway .65)	192.168.10.66	192.168.10.67	192.168.10.68

4. Default Gateway for Each LAN

- Hosts on the Administration LAN (192.168.10.0/26) use default gateway 192.168.10.1 to reach the Finance LAN, or any other network, through the Layer-3 router.
- Hosts on the Finance LAN (192.168.10.64/26) use default gateway 192.168.10.65 for the same purpose.

Both gateway addresses are the router's own interface IPs on each respective LAN—since the two departments sit on different subnets, all inter-department traffic (Administration↔Finance) must be routed through the Layer-3 router via these

gateways rather than being switched directly.